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Studierichting: Informatica

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$$\begin{aligned} & \lim_{x \rightarrow \infty} \frac{x - \sqrt{x^2 - 1}}{3x - 1} \\ &= \lim_{x \rightarrow \infty} \frac{x - \sqrt{x^2 - 1}}{3x - 1} \cdot \frac{x + \sqrt{x^2 - 1}}{x + \sqrt{x^2 - 1}} \\ &= \lim_{x \rightarrow \infty} \frac{x^2 - (x^2 - 1)}{(3x - 1) \cdot (x + \sqrt{x^2 - 1})} \\ & \text{Als } x \rightarrow +\infty \Rightarrow \lim_{x \rightarrow +\infty} \frac{1}{(3x - 1) \left(1 + \sqrt{1 - \frac{1}{x^2}}\right) x} \\ &= \frac{1}{2} \lim_{x \rightarrow +\infty} \frac{1}{3x^2 - x} \\ &= \frac{1}{6} \lim_{x \rightarrow +\infty} \frac{1}{x^2} \\ &= 0 \\ &= \lim_{x \rightarrow +\infty} \frac{1}{2x} = 0 \\ & \text{Als } x \rightarrow -\infty \Rightarrow \lim_{x \rightarrow -\infty} \frac{x - \sqrt{x^2 - 1}}{3x - 1} \\ &= \frac{\lim_{x \rightarrow -\infty} x - \sqrt{x^2 - 1}}{\lim_{x \rightarrow -\infty} 3x - 1} \\ &= \frac{\lim_{x \rightarrow -\infty} 1 - \frac{\sqrt{x^2 - 1}}{x}}{\lim_{x \rightarrow -\infty} 3 - \frac{1}{x}} \\ &= \frac{\lim_{x \rightarrow -\infty} 1 - \sqrt{1 - \frac{1}{x^2}}}{3} \\ &= \frac{1 - (-\sqrt{1})}{3} \\ &= \frac{2}{3} \end{aligned}$$

References